



IILM UNIVERSITY, GREATER NOIDA

INTERNSHIP REPORT PRESENTATION

AI-Powered Job Seeker

Virtual Internship

Resume – Job Description Matching using NLP | TF-IDF & Cosine Similarity

Suchet Khemraj

Roll No. 25SCS1003005619 | B.Tech CSE

July – August 2026

The Problem

Job seekers apply blind.

They rarely know if their resume actually matches a job description before hitting submit.

ATS systems filter silently.

Applicant Tracking Systems reject strong candidates simply for missing the “right” keywords.

Manual comparison is tedious.

Reading a resume against a job posting line-by-line is slow and subjective.

75%

of resumes are rejected by ATS software before a recruiter ever reads them.

Source: widely reported industry estimate on ATS resume filtering

Objectives

1

Extract & Process

Pull clean text out of PDF/TXT resumes automatically.

2

Score with NLP

Use TF-IDF + Cosine Similarity to compute a resume-JD match score.

3

Find the Gaps

Surface important keywords missing from the resume.

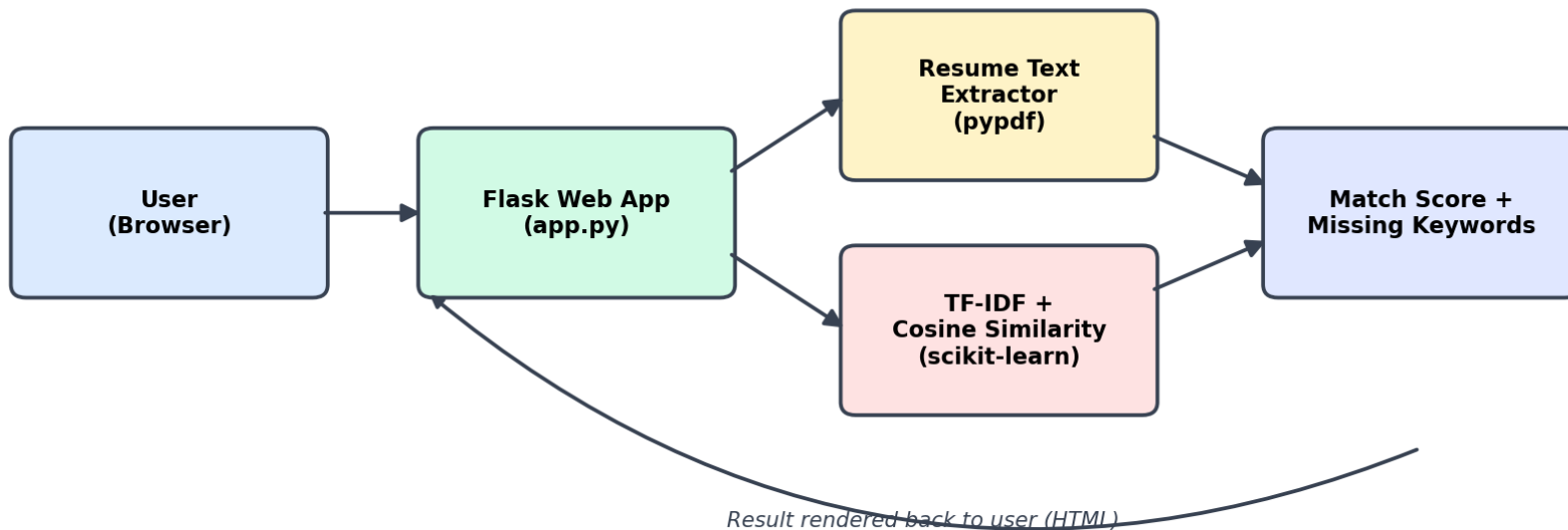
4

Stay Lightweight

Build a simple Flask app — no paid APIs, runs offline.

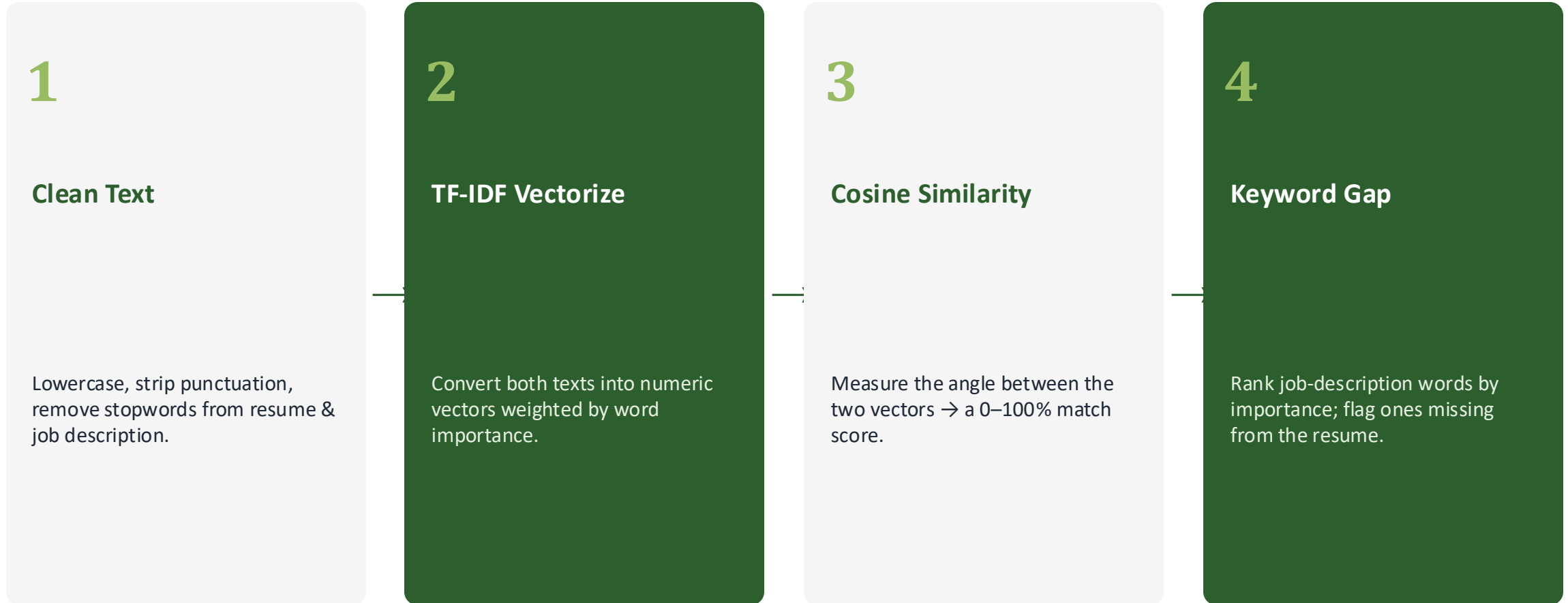
System Architecture

System Architecture — AI-Powered Job Seeker Application



Frontend (HTML) → Flask backend → NLP engine (pypdf + scikit-learn) → results rendered back to the browser.

How the Matching Works



Tech Stack

Python 3

Core language

Flask

Web framework

scikit-learn

TF-IDF + Cosine Similarity

pypdf

PDF text extraction

HTML / CSS

Frontend interface

Live Demo

Application Home Page

AI-Powered Job Seeker Application

Upload your resume and a job description — get an AI-driven match score and see exactly which keywords to add.

Upload Resume (.pdf or .txt)

Choose File No file selected

Paste Job Description

Paste the full job posting text here...

Analyze Match

Live Demo

Match Score & Missing Keywords

AI-Powered Job Seeker Application

Upload your resume and a job description — get an AI-driven match score and see exactly which keywords to add.

Upload Resume (.pdf or .txt)

No file selected

Paste Job Description

Looking for Machine Learning Engineer with Python, scikit-learn, NLP, and Flask experience. SQL knowledge required.

Analyze Match

Resume analyzed: sample_resume.txt

31.56%

Weak Match - resume needs significant tailoring

Keywords from the job posting missing in your resume:

engineer knowledge looking required

Testing Summary

Test Case	Result
Upload valid PDF resume	Pass
Upload valid TXT resume	Pass
No resume uploaded → shows error	Pass
No job description → shows error	Pass
High keyword overlap → high score	Pass
Low keyword overlap → low score	Pass

Conclusion & Future Scope

A lightweight, fully offline NLP tool that gives job seekers an instant, explainable match score — no paid APIs required.

- Semantic matching with Sentence-BERT embeddings
- AI-generated resume improvement suggestions
- Live integration with job-listing APIs (LinkedIn, Indeed)
- User accounts to track match-score history over time



Thank You

AI-Powered Job Seeker Application | B.Tech Major Project

Suchet Khemraj

Roll No. 25SCS1003005619 | IILM University, Greater Noida